

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the factors that alter the birth rate and death rate in human populations	To include food production, advances in medical technology and disease control AND interpretation of demographic charts/graphs. <i>M1.3, M1.5, M3.2, M3.5</i>
(b) the impact of the rise in human population on ecosystems and biodiversity	To include the impacts of humans on the global ecosystem (e.g. impact on abiotic factors such as climate, soils and water quality and on biotic factors such as changes in biodiversity). <i>M1.5</i>
(c) the ecological, economic and scientific importance of species biodiversity	To include aesthetic, medical and agricultural considerations and the use of statistical methods to assess species biodiversity (e.g. Simpson's Diversity Index). <i>M1.5, M2.3</i>
(d) the global food security agenda concerning sustainable food production and food consumption	To include but not limited to food system challenges (food safety, food fraud, food crime and consumer trust) AND the role of the United Nations Food and Agriculture Organisation (to include to FCC (Food Chain Crisis Management Framework) and EMPRES (Emergency Prevention System)).